Standard Course Project Timeline for ME 472 (Winter 2026)

This table outlines the standard project timeline based on the syllabus, integrated with lectures and labs. It focuses on key milestones for the mechatronic design project (team-based, hands-on prototype like a line-following robot or similar system). Remember, this aligns with course objectives #4–#8 (design, component selection, assembly, recommendations, communication). The project is worth 20% of your grade and builds on skills from earlier weeks.

Week #	Dates (2026)	Key Project Activities/Milestones
5	Feb 2–8	**Early Planning and Brainstorming**: Review Mechatronic System Design (De Silva Ch 10). Use discussion board to form teams, brainstorm ideas, outline concepts, and identify contemporary systems (objective #1). Start catalog research for components.
8	Feb 23–Mar 1	**Dedicated Team Work Time**: No lab this week—use for refining design, assigning multi-disciplinary roles (e.g., mechanical, electrical, programming), selecting components from catalogs (objective #5), and initial modeling (sensors/actuators, objective #2).
11	Mar 16–22	**Implementation Phase Start**: Lab 8 – Build H-Bridge Motor Driver for actuators/control. Prototype on breadboard; submit memo. Focus on electromagnetic actuators and basic integration (objective #6).
12	Mar 23–29	**Implementation Continued**: Lab 9 – Add Distance and Touch Sensors. Test closed-loop elements; write basic microcontroller programs (objective #3).

13	Mar 30–Apr 5	**Implementation Continued**: Lab 10 – Integrate Line Sensor. Refine prototype; submit memo. Evaluate benefits/drawbacks (objective #1).
14	Apr 6–12	**Implementation Wrap-Up**: Lab 11 – Implement Speed Control. Full system testing and debugging.
15	Apr 13–19	**Final Integration and Testing**: Lab 12 – System Modeling Lab (De Silva Ch 3). Submit memo. Ensure multi-disciplinary synthesis and recommend solutions (objective #7).
Final	Apr 24 (6:30–9:30 PM)	**Submission and Presentation**: Deliver report, brief presentation, and defend design (objective #8). In-person with written exam.

Notes: This spans ~10 weeks, allowing time for iterative prototyping. Use office hours for feedback. Discussion boards in Weeks 5, 7, 11, 13, and 15 may include project updates.

Accelerated Timeline (Target: Completion by End of March 2026)

If your partner wants the project done by the end of March (around Week 13, Mar 30–Apr 5), this compresses the standard timeline by overlapping phases and using non-class time aggressively. This assumes you've already started (current date: Feb 4, Week 5) and can dedicate extra hours outside labs (e.g., weekends, no-lab weeks). Aim for a functional prototype by Mar 31 to allow buffer for testing/reporting. Coordinate with your team and professor early—accelerating risks missing depth in modeling/control (objectives #2–3). Still submit officially on Apr 24, but treat this as an internal goal for early completion.

Week #	Dates (2026)	Key Project Activities/Milestones (Accelerated)
5	Feb 2–8	**Accelerated Planning**: Brainstorm and finalize ideas via discussion board. Form team, assign roles, and select initial components (objective #5). Complete basic design outline by week-end.
6–7	Feb 9–22	**Early Prototyping Prep**: Use lab time (Labs 5–6: Digital Logic, EM/Microcontroller) to test core components early. Model sensors/actuators (objective #2) outside class; order parts if needed.
8	Feb 23–Mar 1	**Intensive Design and Initial Build**: No lab—dedicate to full design refinement and basic breadboard assembly (objective #6). Integrate microcontroller programming (objective #3); aim for partial prototype.
9	Mar 2–8	**Spring Break Acceleration**: No classes—use for rapid prototyping. Build/test H-Bridge and basic actuators (adapt Lab 8 elements early). Evaluate drawbacks and iterate.

10	Mar 9–15	**Core Implementation**: During Lab 7 (Festo PLC Lab), parallelize project work. Add sensors (distance/touch/line) and basic control systems (objective #2c).
11	Mar 16–22	**Advanced Integration**: Lab 8 – Finalize motor driver and sensors. Implement speed control early; write/test programs. Submit memo as progress check.
12	Mar 23–29	**Testing and Refinement**: Adapt Labs 9–10 elements. Full system testing, modeling (De Silva Ch 3), and debugging. Recommend solutions (objective #7).
13	Mar 30–Apr 5	**Early Completion and Documentation**: Final integration/testing by Mar 31. Draft report and practice presentation (objective #8). Use discussion board for feedback. Buffer for revisions.

Notes: This shortens to ~8 weeks by front-loading (e.g., use Spring Break productively) and overlapping labs with project work. Risks: Less time for iteration—focus on a simpler design to ensure quality. Track progress weekly; if struggling, revert to standard. Discuss feasibility with Prof. Pannier.

ME 472 Project Deliverables Checklist (Winter 2026)

Discussion Board Updates

- [ ] **Week 5:** Team formation, initial brainstorming, and conceptual outline.
- [ ] **Week 7:** Progress update and design refinement.
- [ ] **Week 11:** Implementation update (H-Bridge and motor control).
- [ ] **Week 13:** Integration update (Sensors and prototype status).
- [ ] **Week 15:** Final system modeling and testing summary.

Technical Memos

- [ ] **Lab 8 Memo:** H-Bridge Motor Driver (Circuitry, actuators, and initial control).
- [ ] **Lab 10 Memo:** Line Sensor Integration (Prototype refinement and evaluation).
- [ ] **Lab 12 Memo:** System Modeling (Multi-disciplinary synthesis and modeling results).

Final Project Submission (Due April 24)

- [ ] **Written Report:** Comprehensive documentation including design, component selection, assembly, and recommended solutions.
- [ ] **Project Presentation:** Brief overview and demonstration of the prototype.
- [ ] **Design Defense:** Oral defense of mechatronic system choices and performance.
- [ ] **Prototype Delivery:** Final functional system (e.g., line-following robot).

Key Internal Milestones (Accelerated Goal)

- [ ] **Feb 8:** Roles assigned and core components selected.
- [ ] **Mar 1:** Initial breadboard assembly and basic programming completed.
- [ ] **Mar 8:** Rapid prototyping and testing of core actuators (Spring Break goal).
- [ ] **Mar 22:** Speed control and advanced sensor integration finished.
- [ ] **Mar 31: Target Completion Date** (Full functional testing and debugging).
- [ ] **Apr 5:** Final report draft and presentation rehearsal.

Project Management Tips

- **Version Control:** Maintain a shared folder for all CAD files, microcontroller code, and report drafts to avoid data loss.
- **Early Component Testing:** Test sensors and actuators individually before full system integration to simplify debugging.
- **Documentation:** Take photos and videos of the prototype during different phases; these are valuable for the final report and presentation.